
PRACTICAL WEEK 7

1. Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Create a file:

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano practical7.1.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ |
```

Write the code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 100;
int global_uninitialized;
static int static_global = 200;

void sample_function()
{
    printf("This is a function in the code segment.\n");
}

int main()
{
    int stack_var = 10;
    static int static_local = 20;
    int *heap_var = (int *)malloc(sizeof(int));
    if (heap_var == NULL)
    {
        perror("malloc");
        return 1;
    }
    *heap_var = 30;
    printf("\n===== PROCESS MEMORY ADDRESSES =====\n\n");
    printf("Code / Function address      : %p\n",
           (void *)sample_function);
    printf("Global variable address          : %p\n",
           (void *)&global_var);
    printf("Global uninitialized address : %p\n",
           (void *)&global_uninitialized);
    printf("Static global address            : %p\n",
           (void *)&static_global);
    printf("Static local address             : %p\n",
           (void *)&static_local);
```

```

printf("Heap variable address      : %p\n",
      (void *)heap_var);

printf("Stack variable address     : %p\n",
      (void *)&stack_var);

printf("\nProcess ID (PID)         : %d\n", getpid());

printf("\nProgram is sleeping for 60 seconds.\n");
printf("Use this PID to inspect /proc/<PID>/maps\n");

sleep(60);

free(heap_var);

return 0;
}

```

Run the code:

```

vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano practical7.1.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc practical7.1.c -o practical7.1
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ ./practical7.1

```

OUTPUT:

```

vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano practical7.1.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc practical7.1.c -o practical7.1
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ ./practical7.1

===== PROCESS MEMORY ADDRESSES =====

Code / Function address      : 0x646ca0737229
Global variable address      : 0x646ca073a010
Global uninitialized address : 0x646ca073a020
Static global address        : 0x646ca073a014
Static local address         : 0x646ca073a018
Heap variable address         : 0x646cda3f2010
Stack variable address        : 0x7ffd7d3e8bdc

Process ID (PID)             : 719

Program is sleeping for 60 seconds.
Use this PID to inspect /proc/<PID>/maps
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ |

```

2. Make use of the /proc//maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

Ans:

Go to the file practical7.1 and run it by using gcc practical7.1.c -o practical7.1 and ./practical7.1 then note the PID number then run this in terminal 2:

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ cat /proc/876/maps|
```

OUTPUT:

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ cat /proc/947/maps
5b587aac3000-5b587aac4000 r--p 00000000 08:30 45008 /home/vishwa_teja_veerandi/practical7.1
5b587aac4000-5b587aac5000 r-xp 00001000 08:30 45008 /home/vishwa_teja_veerandi/practical7.1
5b587aac5000-5b587aac6000 r--p 00002000 08:30 45008 /home/vishwa_teja_veerandi/practical7.1
5b587aac6000-5b587aac7000 r--p 00002000 08:30 45008 /home/vishwa_teja_veerandi/practical7.1
5b587aac7000-5b587aac8000 rw-p 00003000 08:30 45008 /home/vishwa_teja_veerandi/practical7.1
5b589e697000-5b589e6b8000 rw-p 00000000 00:00 0 [heap]
77eb79400000-77eb79428000 r--p 00000000 08:30 14176 /usr/lib/x86_64-linux-gnu/libc.so.6
77eb79428000-77eb795c0000 r-xp 00028000 08:30 14176 /usr/lib/x86_64-linux-gnu/libc.so.6
77eb795c0000-77eb7960e000 r--p 001c0000 08:30 14176 /usr/lib/x86_64-linux-gnu/libc.so.6
77eb7960e000-77eb79612000 r--p 0020d000 08:30 14176 /usr/lib/x86_64-linux-gnu/libc.so.6
77eb79612000-77eb79614000 rw-p 00211000 08:30 14176 /usr/lib/x86_64-linux-gnu/libc.so.6
77eb79614000-77eb79621000 rw-p 00000000 00:00 0
77eb797df000-77eb797e2000 rw-p 00000000 00:00 0
77eb797e7000-77eb797e9000 rw-p 00000000 00:00 0
77eb797e9000-77eb797ed000 r--p 00000000 00:00 0
77eb797ed000-77eb797ef000 r--p 00000000 00:00 0
77eb797ef000-77eb797f1000 r-xp 00000000 00:00 0
77eb797f1000-77eb797f2000 r--p 00000000 08:30 13931 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
77eb797f2000-77eb79821000 r-xp 00001000 08:30 13931 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
77eb79821000-77eb7982c000 r--p 00030000 08:30 13931 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
77eb7982c000-77eb7982e000 r--p 0003b000 08:30 13931 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
77eb7982e000-77eb7982f000 rw-p 0003d000 08:30 13931 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
77eb7982f000-77eb79830000 rw-p 00000000 00:00 0
7ffe29f51000-7ffe29f72000 rw-p 00000000 00:00 0 [stack]
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ |
```